

Local Variables and Bindings

CS 350

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Overview: Variables

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| <number>  
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 - `let1` for defining variables
 - Called `let1` because we can only define 1 variable at a time

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 - Will shadow any previous definitions of `x` inside `body`

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```
(define-type ExpS
  ...
  (varS [x : Symbol])
  (let1S [var : Symbol]
        [val : ExpS]
        [body : ExpS]))
```

```
(define-type Exp
  ...
  (varE [x : Symbol])
  (let1E [var : Symbol]
        [val : Exp]
        [body : Exp]))
```

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Desugaring

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```
;; Desugar Expressions into Core Syntax  
(define (desugar [es : ExpS]) : Exp  
  (type-case ExpS es  
    ...  
    [(varS x)  
     (varE x)]  
    [(let1S var val body)  
     (let1E var (desugar val) (desugar body))]  
    ))
```

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(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(let1E var val body)
     (let* ([valV (interp val)]
              [subbedBody TODO])
              (interp subbedBody))])
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- Evaluate the value for the variable var

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- Evaluate the value for the variable `var`
- Replace `var` with that value in `body`
 - Need a new helper function to do this!
- Then, evaluate the *body after the substitution*

Substitution

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- “Replace all occurrences of the variable x with the expression s inside of the expression t ”
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- Critical operation in programming languages

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- Literal: just return the literal unchanged
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- `let` expressions: replace in the body and the value
 - **unless** the bound variable is the variable we're replacing
 - Then, only substitute in the value
 - since the variable in the body is a “new” shadowing variable
- Everything else: recursively substitute in the sub-expressions

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```
(define (subst [toReplace : Symbol]
              [replacedBy : Exp]
              [e : Exp] ) : Exp
  (type-case Exp e
    [(varE x)
     (if (symbol=? x toReplace)
          replacedBy
          (varE x))]
    [(numE n)
     (numE n)]
    [(boolE b)
     (boolE b)]
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              [replacedBy : Exp]
              [e : Exp])
  : Exp
  (type-case Exp e
    ...
    [(let1E var val body)
     (let1E var
       (subst toReplace replacedBy val)
       (if (symbol=? var toReplace)
           body
           (subst toReplace replacedBy body)))]])
```

Code for Substitution: Recursive Cases

- Just recursively substitute in the parts

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```
(define (subst [toReplace : Symbol]
               [replacedBy : Exp]
               [e : Exp])
  : Exp
  (type-case Exp e
    ...
    [(plusE l r)
     (plusE (subst toReplace replacedBy l)
              (subst toReplace replacedBy r))]
    [(timesE l r)
     (timesE (subst toReplace replacedBy l)
               (subst toReplace replacedBy r))]
    [(cndE test thn els)
     (cndE (subst toReplace replacedBy test)
            (subst toReplace replacedBy thn)
            (subst toReplace replacedBy els))]
    [(zero?E e)
     (zero?E (subst toReplace replacedBy e))])
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(define (Value->Exp [v : Value])  
  : Exp  
  (type-case Value v  
    [(boolV b)  
     (boolE b)]  
    [(numV n)  
     (numE n)]))
```

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```
(define (interp [e : Exp] ) : Value
  (type-case Exp e
    ...
    [(varE x)
     (error 'interp
            (string-append "Undefined variable: "
                           (symbol->string x)))]
```